1.2 Modelling Relationships using Graphs

We may wish to explore the *relationships* between two or more variables, such as when considering:

Questions

- How will the **money** in my savings account grow over time if I make regular payments?
- What might a car's **braking distance** by when travelling at different speeds?
- How might the distance travelled in a taxi be related to the **fare**?

We should aim to identify where possible for any relationship:

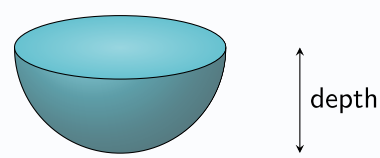
- The independent variable - commonly either *time* or one which we control.
- The **depdendent** variable - which changes in **response** to the other.

Creating graph of a possible model is a good starting point in understanding any relationship. The *independent* variable should be placed along the (horizontal) x-axis and the *response* variable placed on the (vertical) y-axis.

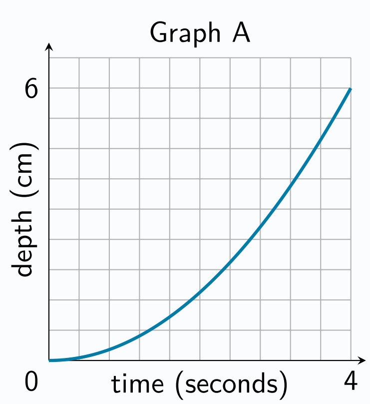
You may have to *sketch* a possible graph of a relationship, or *select from a choice of graphs*, giving an explanation.

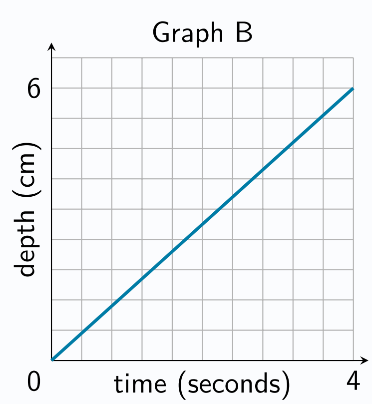
Example

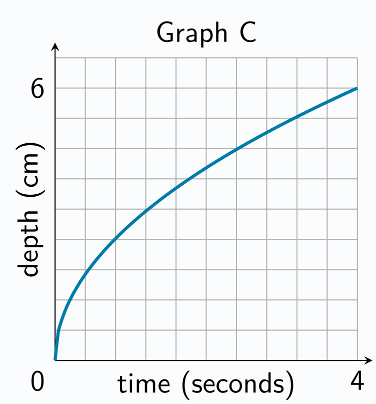
Problem:
Bob pours water into an empty cup at a constant rate.
The interior of the cup is in the shape of a hemisphere, as shown below.



State which of graphs A, B or C below might be best suited to model the relationship between the time since Bob began pouring the water and the depth of water in the cup, explaining your answer.







Solution:
Graph C, since the depth will increase more quickly at the start, then slow down towards the end.